# THE ONLY FIXED POINTS OF $x \mapsto \mathrm{rev}(x) - \mathrm{maxdig}(x)$ ARE 0 AND 89: EARLS' CONJECTURE IN OEIS A097154

MATH0

Abstract. For a nonnegative integer $x$ let $\mathrm{rev}(x)$ be the number obtained by reversing the decimal digits of $x$, and let $\mathrm{maxdig}(x)$ be the largest decimal digit of $x$. We prove that the map $f(x) = \mathrm{rev}(x) - \mathrm{maxdig}(x)$ has exactly two fixed points, 0 and 89, which was stated as a conjecture by Jason Earls in OEIS A097154 (2004) and repeated in A097155 and A097157. The proof combines the congruence $\mathrm{rev}(x) \equiv x \pmod 9$, which forces $\mathrm{maxdig}(x) = 9$ at a nonzero fixed point, with a comparison of the leading and units digits. To our knowledge the result is new, although the argument is elementary; the OEIS entries still list the statement as a conjecture. The theorem is verified in Lean 4 against a statement that math0 formalized from the OEIS entry, reviewed for fidelity and frozen before any proof attempt, whose definitions are checked by the kernel against the listed OEIS terms; this statement is not part of the formal-conjectures repository.

## 1. Introduction

In August 2004 Jason Earls contributed to the OEIS a group of sequences A097153–A097158 about the map

$$f(x) = \mathrm{rev}(x) - \mathrm{maxdig}(x) \qquad (x \geq 0),$$

where $\mathrm{rev}(x)$ is the decimal reversal of $x$ (sequence A004086 [4], with leading zeros of the reversed string dropped, so that $\mathrm{rev}(120) = 21$) and $\mathrm{maxdig}(x)$ is the largest decimal digit of $x$. The values $f(n)$ form A097153 [5]. A097154 [6] gives the number of iterations of $f$ needed to bring $n$ to 0, or $-1$ if 0 is never reached; for example $12 \mapsto 19 \mapsto 82 \mapsto 20 \mapsto 0$, so that $a(12) = 4$. The first value $-1$ is $a(89)$, because $f(89) = 98 - 9 = 89$. A097155 [7] lists the numbers whose orbit reaches 89, A097157 [9] the number of iterations they need to reach 89, A097156 [8] the least number that needs exactly $n$ iterations to reach 0, and A097158 [10] the numbers that set a new record for the number of iterations needed to reach 0. The comments of A097154, A097155 and A097157 all contain the sentence "It is conjectured that 0 and 89 are the only fixed points of f." We prove this (Theorem 2).

**Method.** If $f(x) = x$ then $\mathrm{rev}(x) = x + \mathrm{maxdig}(x)$. Reversal preserves the digit sum, so $\mathrm{rev}(x) \equiv x \pmod 9$, and therefore $9 \mid \mathrm{maxdig}(x)$; for $x \neq 0$ this gives $\mathrm{maxdig}(x) = 9$. It remains to solve $\mathrm{rev}(x) = x + 9$ with largest digit 9. Writing $x$ as units digit $b$, middle block $M$ and leading digit $a$, the equation modulo 10 leaves two cases for $(a, b)$; one is impossible by size, and the other forces the middle block to be empty, so that $x = 10a + (a + 1)$, and the condition on the largest digit gives $x = 89$.

**Status and prior work.** The local copy of the OEIS data used by math0, which the referees cross-checked against the current OEIS data files, has A097154 at revision #4 (December 15, 2017), A097155 at revision #11 (August 29, 2025) and A097157 at revision #3 (December 15, 2017). All three still state the conjecture, and none has a comment, formula, link or edit note recording a proof or a counterexample; the only edit note in A097155 is a correction of its data

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by Harvey P. Dale (2011). The related entries A097153, A097156 and A097158, with extensions by Robert G. Wilson v and Klaus Brockhaus, are computational, and no other OEIS entry in that copy refers to A097153–A097158. Web searches by the math0 referees (for the definition of the map, for phrasings such as "reversal minus largest digit", and in the SeqFan archives and question-and-answer sites) found no proof; they found only the OEIS pages themselves and work on other digit-reversal maps. The statement is not among the OEIS statements of the formal-conjectures repository [1]. We therefore classify the result as *new*. The argument uses only congruences modulo 9 and 10 and a size comparison, and it may well have been noticed without being recorded.

We note that the conjecture is about fixed points in the literal sense, $f(x) = x$; A097155 calls 89 "the fixed point 89". We do not address the stronger question whether $f$ has periodic orbits of length at least 2, that is, whether every orbit ends at 0 or at 89 (Remark 9).

## 2. Statement

**Notation.** For a finite list $\ell = (e_0, \dots, e_{n-1})$ of natural numbers we write $|\ell| = n$,

$$\mathrm{val}(\ell) = \sum_{i=0}^{n-1} e_i \, 10^i, \qquad \ell^R = (e_{n-1}, \dots, e_0), \qquad \mu(\ell) = \max(0, e_0, \dots, e_{n-1}).$$

In Lean these are `Nat.ofDigits 10` $\ell$, $\ell$`.reverse` and $\ell$`.foldr max 0`. We use $\mathrm{val}(()) = 0$, $\mathrm{val}(e :: \ell) = e + 10\,\mathrm{val}(\ell)$ and $\mathrm{val}(\ell +\!\!+ \ell') = \mathrm{val}(\ell) + 10^{|\ell|}\,\mathrm{val}(\ell')$, where $e :: \ell$ is the list with first entry $e$ followed by $\ell$ and $+\!\!+$ is concatenation. For $x \in \mathbb{N}$ let $\delta(x)$ be the list of decimal digits of $x$, least significant first (`Nat.digits 10 x`). Thus $\delta(0) = ()$, and for $x \geq 1$ the list $\delta(x)$ is nonempty, its entries lie in $\{0, \dots, 9\}$, its last entry (the leading digit) is nonzero, and $\mathrm{val}(\delta(x)) = x$.

**Definition 1.** For $x \in \mathbb{N}$ put

$$\mathrm{rev}(x) = \mathrm{val}\big(\delta(x)^R\big), \qquad \mathrm{maxdig}(x) = \mu\big(\delta(x)\big), \qquad f(x) = \mathrm{rev}(x) - \mathrm{maxdig}(x).$$

So $\mathrm{rev}(x)$ is the number whose decimal digits are those of $x$ in reverse order, with leading zeros dropped, and $\mathrm{rev}(0) = \mathrm{maxdig}(0) = 0$. By Lemma 4(c) below, $\mathrm{maxdig}(x) \leq \mathrm{rev}(x)$, so $f$ maps $\mathbb{N}$ to $\mathbb{N}$ and the subtraction is exact. For example $f(10) = 1 - 1 = 0$ and $f(89) = 98 - 9 = 89$.

**Theorem 2.** *For every $x \in \mathbb{N}$, $f(x) = x$ if and only if $x = 0$ or $x = 89$.*

**The OEIS conjecture.** The entry A097154 [6], contributed by Jason Earls on August 1, 2004, is named "Number of iterations of f(x) = reverse(x) - maxdigit(x) needed to reach zero, or -1 if zero is never reached, where maxdig(n) is the largest digit of n", with offset 1 and keywords `base`, `easy`, `sign`. Its COMMENTS section reads, verbatim (the comment carries no separate signature):

```
a(89)=-1 because 89 is the first number that will not reach zero:
98-9=89.  It is conjectured that 0 and 89 are the only fixed points of
f.  See A097155 for numbers that reach 89.
```

The same conjecture appears in the comments of A097155 and A097157, also contributed by Earls on August 1, 2004.

**Lean formalization.** The statement was formalized by math0 in the file `lean/Math0/Statements/OEIS/A097154.lean`. It was reviewed for fidelity and frozen before any proof attempt; its SHA-256 hash is `1d7d144b...cbb8`. The definitions of the map and the statement read as follows (verbatim; the import, docstrings, blank lines and all other declarations are omitted):

```
namespace Math0.Statements.A097154
def rev (x : ℕ) : ℕ := Nat.ofDigits 10 (Nat.digits 10 x).reverse
def maxDigit (x : ℕ) : ℕ := (Nat.digits 10 x).foldr max 0
def f (x : ℕ) : ℕ := rev x - maxDigit x
def statement : Prop :=
  ∀ x : ℕ, f x = x ↔ x = 0 ∨ x = 89
end Math0.Statements.A097154
```

These are the definitions rev, maxdig and $f$ of Definition 1, and `statement` is Theorem 2; the biconditional says both that 0 and 89 are fixed points and that there are no others. The subtraction in `f` is truncated subtraction on $\mathbb{N}$, but the file proves that it never truncates, and the same file also encodes the sequence A097154 itself. Its other declarations include the following (statements shown, proofs omitted):

```
theorem maxDigit_le_rev (x : ℕ) : maxDigit x ≤ rev x
theorem f_eq_int (x : ℕ) : (f x : ℤ) = (rev x : ℤ) - (maxDigit x : ℤ)
theorem f_examples :
    f 12 = 19 ∧ f 19 = 82 ∧ f 82 = 20 ∧ f 20 = 0 ∧ f 0 = 0 ∧ rev 89 = 98 ∧ maxDigit 89 = 9 ∧
      f 89 = 89
def search : ℕ → ℕ → ℕ → ℤ
  | x, k, 0 => if x = 0 then (k : ℤ) else -1
  | x, k, fuel + 1 => if x = 0 then (k : ℤ) else search (f x) (k + 1) fuel
def a (n : ℕ) : ℤ := search n 0 (10 ^ (Nat.digits 10 n).length)
open Classical in
theorem a_spec (n : ℕ) : a n = if h : ∃ k, f^[k] n = 0 then (Nat.find h : ℤ) else -1
```

By `maxDigit_le_rev` and `f_eq_int`, `f` agrees with the integer-valued map of the OEIS entry, so the truncated subtraction creates no junk values. The sequence takes the value $-1$, so `a` has values in $\mathbb{Z}$. It follows the orbit of $n$ for at most $10^d$ steps, where $d$ is the number of digits of $n$, which makes it computable; `a_spec` shows that this bound loses nothing, so that `a` is exactly the OEIS definition (the least $k$ with $f^k(n) = 0$, and $-1$ if there is none). Its proof shows that $f$ maps $\{0, \dots, 10^d - 1\}$ into itself and that the iterates before the first 0 are pairwise distinct, and then applies the pigeonhole principle. Two lemmas proved by `decide +kernel` compare the definitions with the entry: `a_data` checks all 127 listed terms $a(1), \dots, a(127)$ of A097154, including $a(89) = -1$, and `f_examples` checks the example $12 \mapsto 19 \mapsto 82 \mapsto 20 \mapsto 0$ of the entry, together with $f(0) = 0$ and $f(89) = 98 - 9 = 89$. The reversal convention (trailing zeros of $x$ disappear) is the one the data require, since $a(10) = a(20) = \cdots = 1$. The statement quantifies over all $x \in \mathbb{N}$, including $x = 0$, which the conjecture names as a fixed point, and adds no hypotheses.

## 3. Proof

The proof follows the Lean file `lean/Math0/Frontier/OEIS/A097154.lean`; the name of the corresponding Lean declaration is given with each step. Lemma 3 and parts (b) and (c) of Lemma 4 are proved in the statement file and reused; part (a) of Lemma 4 is proved in the proof file.

**Lemma 3** (`le_ofDigits_of_mem`)**.** *If $e$ is an entry of the list $\ell$, then $e \le \mathrm{val}(\ell)$.*

*Proof.* By induction on $\ell$. The empty list has no entries. If $\ell = e' :: \ell'$, then $\mathrm{val}(\ell) = e' + 10\,\mathrm{val}(\ell')$; either $e = e' \le \mathrm{val}(\ell)$, or $e$ is an entry of $\ell'$ and $e \le \mathrm{val}(\ell') \le \mathrm{val}(\ell)$ by induction. □

**Lemma 4** (`le_foldr_max`, `foldr_max_le`, `maxDigit_le_rev`)**.** *Let $\ell$ be a list of natural numbers.*

(a) *Every entry $e$ of $\ell$ satisfies $e \le \mu(\ell)$.*
(b) *If $c \in \mathbb{N}$ and every entry of $\ell$ is at most $c$, then $\mu(\ell) \le c$.*
(c) *For every $x \in \mathbb{N}$, $\mathrm{maxdig}(x) \le \mathrm{rev}(x)$.*

*Proof.* Parts (a) and (b) follow by induction on $\ell$ from $\mu(()) = 0$ and $\mu(e :: \ell) = \max(e, \mu(\ell))$. For (c), every entry of $\delta(x)$ is an entry of $\delta(x)^R$, hence at most $\mathrm{val}(\delta(x)^R) = \mathrm{rev}(x)$ by Lemma 3; now apply (b) with $c = \mathrm{rev}(x)$. □

**Lemma 5** (`ofDigits_reverse_mod`)**.** *For every list $\ell$ of natural numbers,* $\mathrm{val}(\ell^R) \equiv \mathrm{val}(\ell) \pmod 9$.

*Proof.* Since $10 \equiv 1 \pmod 9$, we have $\mathrm{val}(\ell) = \sum_i e_i 10^i \equiv \sum_i e_i \pmod 9$ (in Lean, `Nat.ofDigits_modEq` and `Nat.ofDigits_one`), and likewise for $\ell^R$. The entries of $\ell$ and $\ell^R$ have the same sum (`List.sum_reverse`). □

**Lemma 6** (`ofDigits_eq_zero`)**.** *If every entry of $\ell$ is $0$, then* $\mathrm{val}(\ell) = 0$.

*Proof.* Immediate by induction on $\ell$. □

We also use that $\mathrm{val}(\ell) < 10^{|\ell|}$ whenever all entries of $\ell$ are at most 9 (`Nat.ofDigits_lt_base_pow_length` in Mathlib). The heart of the proof is a statement about digit lists.

**Proposition 7** (`key`)**.** *Let $\ell$ be a nonempty list with entries in $\{0, \dots, 9\}$ whose last entry is nonzero, and suppose that*

$$\mathrm{val}(\ell^R) = \mathrm{val}(\ell) + \mu(\ell). \tag{1}$$

*Then $\ell = (9, 8)$.*

*Proof.* Put $c = \mu(\ell)$. All entries of $\ell$ are at most 9, so $c \le 9$ by Lemma 4(b).

*One entry.* If $\ell = (b)$, then $\ell^R = \ell$, so (1) gives $c = 0$. By Lemma 4(a), $b \le c = 0$, contradicting the assumption that the last entry $b$ is nonzero.

*At least two entries.* Then $\ell = b :: (M +\!\!+ (a))$ (`List.eq_nil_or_concat'`), where $b$ is the first entry (the units digit), $a$ is the last entry (the leading digit) and $M$ is a possibly empty list of length $k$. We have $1 \le a \le 9$, $0 \le b \le 9$, and $a \le c$, $b \le c$ by Lemma 4(a). Put

$$m = \mathrm{val}(M), \qquad r = \mathrm{val}(M^R), \qquad P = 10^k.$$

Since $\ell^R = a :: (M^R +\!\!+ (b))$ and $|M^R| = k$, the rules for val give

$$\mathrm{val}(\ell) = b + 10m + 10Pa, \qquad \mathrm{val}(\ell^R) = a + 10r + 10Pb. \tag{2}$$

We record three facts.

(i) $m < P$ and $r < P$, because the entries of $M$ and of $M^R$ are at most 9.
(ii) $P \ge 1$, and $P \ge 10$ if $k \ge 1$.
(iii) If $r = 0$ then $m = 0$. Indeed, each entry of $M$ is an entry of $M^R$, hence at most $r = 0$ by Lemma 3, and Lemma 6 gives $m = 0$.

By (2), the hypothesis (1) reads

$$a + 10r + 10Pb = b + 10m + 10Pa + c. \tag{3}$$

*Step 1: $c = 9$.* By Lemma 5, $c = \mathrm{val}(\ell^R) - \mathrm{val}(\ell) \equiv 0 \pmod 9$. Since $1 \le a \le c \le 9$, we get $c = 9$.

*Step 2: $b = a + 1$, or $a = 9$ and $b = 0$.* Reducing (3) modulo 10 gives $a \equiv b + 9 \pmod{10}$. If $b \ge 1$ then $a \equiv b - 1 \pmod{10}$ with $0 \le a$, $b - 1 \le 9$, so $a = b - 1$, that is, $b = a + 1$. If $b = 0$ then $a \equiv 9 \pmod{10}$, so $a = 9$.

*Step 3: the case $a = 9$, $b = 0$ is impossible.* Then (3) becomes $9 + 10r = 10m + 90P + 9$, that is $r = m + 9P \ge P$, contradicting (i).

*Step 4: the case $b = a + 1$ gives $\ell = (9, 8)$.* Then (3) becomes $a + 10r + 10Pa + 10P = a + 1 + 10m + 10Pa + 9$, that is,

$$r + P = m + 1.$$

By (i), $m + 1 \le P$, so $r = 0$. By (iii), $m = 0$, so $P = 1$, and by (ii), $k = 0$; thus $M$ is empty and $\ell = (a + 1, a)$. Both entries of $\ell$ are at most $a + 1$, so $9 = c \le a + 1$ by Lemma 4(b), while $a + 1 = b \le 9$. Hence $a = 8$ and $\ell = (9, 8)$. □

In Lean, $m$, $r$ and $P$ are replaced by variables (`generalize`) after (i)–(iii) and (2) have been established, and the arithmetic in Steps 1–4 is done by `omega`, a decision procedure for linear arithmetic over $\mathbb{N}$ and $\mathbb{Z}$; the products $Pa$ and $Pb$ are treated as atoms, and the identity $P(a + 1) = Pa + P$ is supplied by `ring`. The bound $c \le a + 1$ at the end of Step 4 comes from Lemma 4(b) (`foldr_max_le`).

*Proof of Theorem 2 (`result`).* If $x = 0$ or $x = 89$, then $f(x) = x$ by direct computation: $f(0) = 0 - 0 = 0$ and $f(89) = 98 - 9 = 89$ (these are two conjuncts of `f_examples`, proved by evaluation in the kernel).

Conversely, let $f(x) = x$ and suppose $x \neq 0$. By Lemma 4(c) the subtraction in $f(x)$ is exact, so $\mathrm{rev}(x) = x + \mathrm{maxdig}(x)$. Apply Proposition 7 to $\ell = \delta(x)$: it is nonempty because $x \neq 0$ (`Nat.digits_ne_nil_iff_ne_zero`), its entries are at most 9 (`Nat.digits_lt_base`), its last entry is nonzero (`Nat.getLast_digit_ne_zero`), and (1) holds because $\mathrm{val}(\delta(x)) = x$ (`Nat.ofDigits_digits`). Hence $\delta(x) = (9, 8)$, and $x = \mathrm{val}(\delta(x)) = 9 + 10 \cdot 8 = 89$. □

**Remark 8** (a by-product; not formalized)**.** In Steps 2–4 the hypothesis (1) enters only through (3) with $c = 9$, except in the last sentence of Step 4. Running the same argument for the equation $\mathrm{rev}(x) = x + 9$ with $x \ge 1$ shows that its solutions are exactly $x = 11a + 1$ with $1 \le a \le 8$, that is $12, 23, 34, 45, 56, 67, 78, 89$; conversely $\mathrm{rev}(11a + 1) = 10(a + 1) + a = (11a + 1) + 9$ for these $a$. Step 1 shows that every fixed point $x \neq 0$ has $\mathrm{maxdig}(x) = 9$, and among these eight numbers only 89 has largest digit 9. (The two-digit case of the equation $\mathrm{rev}(x) = x + 9$ is a familiar puzzle; the point here is only that no longer solutions exist.) This remark is not part of the Lean development; we checked the list of solutions numerically for $x < 10^6$.

**Remark 9** (periodic orbits; not addressed)**.** Since $f$ maps $\{0, \dots, 10^d - 1\}$ into itself (this is the lemma `f_lt` of the statement file), every orbit of $f$ is eventually periodic. Theorem 2 determines the periodic orbits of length 1. We have checked by computer that every orbit starting below $10^6$ ends at 0 or at 89, but we do not know whether $f$ has periodic orbits of length at least 2; this question is not part of the conjecture.

## 4. Machine verification

**Files and declarations.** The proof is in the file `lean/Math0/Frontier/OEIS/A097154.lean` of the math0 repository, `https://github.com/Foundation-0/math0`. It imports the frozen statement file `lean/Math0/Statements/OEIS/A097154.lean` (Section 2) and the audit commands in `lean/Math0/Audit.lean`. The main declaration is `Math0.Frontier.A097154.result`, of type `Math0.Statements.A097154.statement`. The helper lemmas `le_foldr_max`, `ofDigits_eq_zero`, `ofDigits_reverse_mod` and `key` (Lemmas 4(a), 6, 5 and Proposition 7) are private; the lemmas `le_ofDigits_of_mem`, `foldr_max_le`, `maxDigit_le_rev` and `f_examples` come from the statement file. The core lemma, the main theorem and the final command read as follows (verbatim; proofs omitted):

```
private theorem key (L : List ℕ) (hL : ∀ d ∈ L, d < 10) (hlast : ∀ h : L ≠ [], L.getLast h ≠ 0)
    (heq : Nat.ofDigits 10 L.reverse = Nat.ofDigits 10 L + L.foldr max 0) (hne : L ≠ []) :
    L = [9, 8]
theorem result : statement
```

```
#math0_settles Math0.Frontier.A097154.result Math0.Statements.A097154.statement
```

Here `key` is Proposition 7, and the proof of `result` is the proof of Theorem 2 given above, step by step. The proof file uses no `native_decide`, `sorry` or `set_option`; its only use of `decide` evaluates $\mathrm{val}((9,8)) = 89$. The lemma `f_examples` in the statement file is proved by `decide +kernel`, which evaluates in the kernel and adds no axioms.

**Audit.** The command `#math0_settles`, defined in `lean/Math0/Audit.lean` for statements that math0 formalizes itself, fails unless (1) `statement` is a definition of type `Prop` without universe parameters; (2) `statement` and `result` use no axioms besides `propext`, `Classical.choice` and `Quot.sound` (this rejects `sorryAx`, `Lean.ofReduceBool` from `native_decide`, and custom axioms); and (3) the type of `result` is syntactically the statement or its negation, up to metadata annotations and without unfolding. Here it reports

```
math0 audit: 'Math0.Statements.A097154.statement' depends on axioms [Classical.choice, Quot.sound, propext]
math0 audit: 'Math0.Frontier.A097154.result' depends on axioms [Classical.choice, Quot.sound, propext]
math0 statement check: 'Math0.Frontier.A097154.result' proves 'Math0.Statements.A097154.statement'
```

**Toolchain and reproduction.** We use Lean 4 [3] `v4.33.1` and Mathlib [2] `v4.33.1`, pinned in `lean/`; the repository also pins formal-conjectures [1] at commit `c77fdd4`, which this result does not use. To reproduce, run `scripts/verify.sh` from the repository root: it fetches the Mathlib build cache, builds all math0 results with `lake build` (re-running every audit command), checks that the pinned `formal_conjectures` and `mathlib` checkouts are unmodified, and reports the number of passed statement checks. The single file can be checked with `lake env lean Math0/Frontier/OEIS/A097154.lean` in the directory `lean`; this takes about twenty seconds on our machine once Mathlib and the statement file have been built.

## References


[1] M. Firsching, P. Lezeau, S. Mercuri, M. Z. Horváth, Y. Dillies, C. Sönne, E. Wieser, F. Zhang, T. Hubert, B. Agüera y Arcas, and P. Kohli, *Formal Conjectures: An Open and Evolving Benchmark for Verified Discovery in Mathematics*, arXiv:2605.13171 [cs.AI], 2026. Repository: `https://github.com/google-deepmind/formal-conjectures`.

[2] The mathlib Community, *The Lean mathematical library*, in: Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), ACM, 2020, 367–381.

[3] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in: Automated Deduction – CADE 28, Lecture Notes in Comput. Sci. **12699**, Springer, 2021, 625–635.

[4] The OEIS Foundation Inc., `https://oeis.org/A004086` (read $n$ backwards), entry by N. J. A. Sloane.

[5] The OEIS Foundation Inc., `https://oeis.org/A097153` (reversal of $n$ minus largest digit of $n$), entry by J. Earls (August 1, 2004).

[6] The OEIS Foundation Inc., `https://oeis.org/A097154`, entry by J. Earls (August 1, 2004); revision #4, December 15, 2017.

[7] The OEIS Foundation Inc., `https://oeis.org/A097155`, entry by J. Earls (August 1, 2004), corrected by H. P. Dale (2011); revision #11, August 29, 2025.

[8] The OEIS Foundation Inc., `https://oeis.org/A097156`, entry by J. Earls (August 1, 2004), more terms by R. G. Wilson v (2004).

[9] The OEIS Foundation Inc., `https://oeis.org/A097157`, entry by J. Earls (August 1, 2004); revision #3, December 15, 2017.

[10] The OEIS Foundation Inc., `https://oeis.org/A097158`, entry by J. Earls (August 1, 2004), terms added by K. Brockhaus (2004).